

Video Transcript

Video 11: Firebase: A Serverless Cloud Database (04:59)

Firebase, as the name indicates, it's a back-end service or a base. And practically what we're going to be doing here is we're going to be using that back end as a database, it's storage, and we're going to be connecting to it from Python. So, as you can see here, I have created a database project, and I am at the console. I'm going to go ahead and click on 'Project Overview' and 'Project Settings.' You can see here some of the information for this project. Now, we're going to need some permissions to connect from Python. And we can get that through service accounts. And if you select here the 'Python' option, you can generate a new private key that will be a file that you can download and save.

And as you can see here on my listing of files, I have done precisely that. I won't share it because I don't want to delete the project afterwards, but all you have to do is simply add it to your project and then as you will see, reference it from the code. So, once you have done that, once you have created your credentials and you have created your project within Firebase, now the next thing to do is to go ahead and add the package that you need from Python. And so, in this case, we're going to use pip3 installer, and we're going to go ahead and install Firebase admin package.

So, as you can see there, and yes, I left out the install. So, I'm going to go ahead and enter 'Install'. And as you can see there, I have made the installation. Now once we do that, then we can import credentials and enter also the database functionality. Now at this point here on line seven, you can see here is where you would reference the path to the credentials that you downloaded from Firebase, the site. And then, you'd need to initialize the app with your service account and, point it to the database that you're working with.

And so here, if we go to the Cloud Firestore, we can see here some of the services that exist. And in this case, we're going to be using the real time database as you can see here. And this is the path that we need. That would also, if you take a look at your credential's files, you'd be able to see that information there as well. And so, once we have that, we can go ahead and continue here on our file. And you can see here that I am referencing, this is a hierarchical database, think of JSON.

And you can see there that I am referencing the root of the information that I'm going to construct. And then I'm going to create a child called users. And to that, we're going to go ahead and add the data that you see here, which is a couple of keys, and you can see there are some of the properties. Now below that I'm simply going to go ahead and read some of that information back. And once I have it, well, actually, no, let me see that. That's going to be an update. And then on the last one, I'm going to go ahead and read that data back and then just write it to the console.

Now, as you can see here, when I was testing, I created all of that information. And what I'm going to do is that I'm going to go ahead and delete it. And this is now gone, as you can see. Now, let's go ahead and go back to the code. And let's go ahead and run. It's going to be python3. Anyway, my file is Fire. And so, as you can see there, I have written into the database, and I am now reading it back. And if we take a look at the real time database through the browser, you can see here that information has in fact been recreated. So, there you have it, a pretty fast and quick way to store information, in this case JSON information onto the database. You can see there that once we installed the package, it was just a matter of setting up the credentials, identifying the database that we're going to write into, then adding some structure for the data that you're pushing into.

You can see here how to update and this block and then after that, we're simply going to read it back and writing it to the console. So, this structure in this database is a pretty friendly way to get started with cloud storage without much structure in being able to pull and retrieve some information.